

Stoichiometry of Chemical Reactions

Answering "How Much?"

Dr. Michael Evans

April 29, 2022

Georgia Institute of Technology

Outline

1. Writing and Balancing Chemical Equations
2. Classifying Chemical Reactions
3. Reaction Stoichiometry
4. Reaction Yields

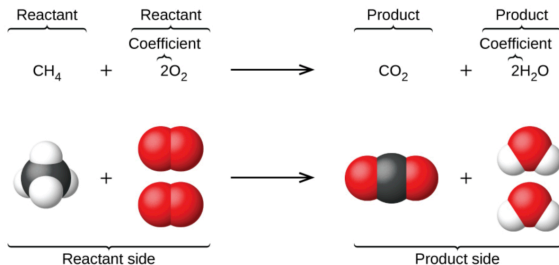
Writing and Balancing Chemical Equations

Chemical equations describe reactions

In a chemical reaction, atoms gain/lose electrons and/or bonds are broken or formed.

Definition

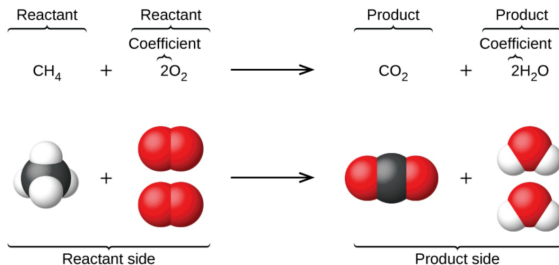
A **chemical equation** shows the identities, numbers, and phases of all species involved in a chemical reaction.



Coefficients are ratios applicable at any scale

Definition

The **stoichiometric coefficient** of a species is the number of particles of that species (atoms, molecules, ions) consumed or produced in each **reaction event**. Coefficients are *relative* numbers that can be applied at molecular or molar scale.



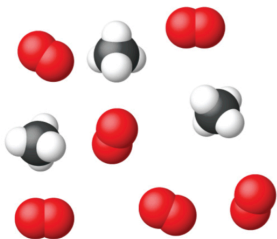
Chemical equations are used to plan reactions

Predicting the Composition of a Product Mixture

Using the balanced chemical equation below, draw the composition of the mixture at the end of the reaction.



Mixture before reaction



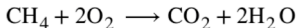
Mixture after reaction



All reasonable chemical equations are balanced

Foundational Skill

In a **balanced** chemical equation, all of the atoms on the reactant side can be found on the product side. You *must* be able to easily balanced equations by inspection and rapidly check whether an equation is balanced! Click here for additional practice.



Element	Reactants	Products	Balanced?
C	$1 \times 1 = 1$	$1 \times 1 = 1$	$1 = 1$, yes
H	$4 \times 1 = 4$	$2 \times 2 = 4$	$4 = 4$, yes
O	$2 \times 2 = 4$	$(1 \times 2) + (2 \times 1) = 4$	$4 = 4$, yes

Figure 1: Check for balance by considering atoms of each element.

Complete chemical equations include indicators of states

- The **state** of a substance is indicated with an abbreviation in parentheses following its formula:
 - (s) Pure solid
 - (*l*) Pure liquid
 - (g) Gas
 - (aq) Solute in aqueous solution
- Special conditions required for the reaction to proceed are indicated above the arrow.
 - Δ Heating required
 - reflux** Heating at the boiling point of the solvent
 - P, T, t* Specific pressures, temperatures, times, etc.

Ionic equations take different forms depending on context

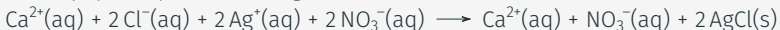
Definition

Molecular equations include ionic compounds in a (misleading) "associated" form involving a full chemical formula. E.g.,



Definition

Complete ionic equations include all dissolved ions as distinct aqueous species (more physically realistic). E.g.,



Definition

Net ionic equations include dissolved ions as distinct aqueous species (as above) but omit **spectator ions**. E.g.,



We can write chemical equations from textual descriptions

Molecular and Ionic Equations

When carbon dioxide is dissolved in an aqueous solution of sodium hydroxide, the mixture reacts to yield aqueous sodium carbonate and liquid water. Write balanced molecular, complete ionic, and net ionic equations for this process. *Include states!*

Classifying Chemical Reactions

Reactions display patterns of chemical change

Classifying Chemical Reactions

We classify chemical reactions according to common patterns of reactivity. By learning the general principles that govern each reaction type, we will be able to make predictions about the products, feasibility, and rates of these reactions.

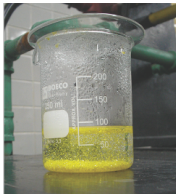


Figure 2: A precipitation reaction.

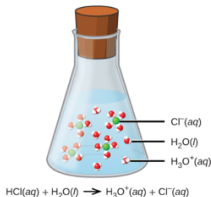


Figure 3: A Brønsted acid-base reaction.

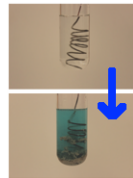


Figure 4: An oxidation-reduction reaction.

Solids form in precipitation reactions

Definition

In a **precipitation reaction**, dissolved substances react to form a solid product: (aq) $\rightarrow$ (s). Many involve the combination of ions in aqueous solution into insoluble solids.

Definition

Dissolution is the formation of a homogeneous phase as solute molecules are surrounded by solvent. **Dissociation** is the separation of ions in an ionic compound as they are individually surrounded by solvent molecules.



Figure 5: Solid sodium chloride dissolves in water *and* its ions dissociate.

Spectator ions are common in precipitation reactions

Writing Complete and Net Ionic Equations

Write balanced complete ionic and net ionic equations for the precipitation reaction below.



Every solute has a solubility limit

Definition

Solubility refers to the maximum amount of a solute that will dissolve in a set amount of solution (typically aqueous). The units of solubility are mass per liter of solution or molarity.

Soluble Solute with a very high solubility value

Insoluble Solute with a very low solubility value (≈ 0)

Sparingly soluble Solute with neither very high nor very low solubility value

Ionic compounds display patterns in solubility

Definition

Solubility rules enable us to predict whether a given ionic compound is soluble in water based on the identity of its ions.

Soluble ionic compounds contain these ions	Exceptions
Group 1 cations: Li^+ , Na^+ , K^+ , Rb^+ , Cs^+	[none]
Cl^- , Br^- , I^-	Salts of Ag^+ , Hg_2^{2+} , and Pb^{2+}
F^-	Salts of group 2 cations, Pb^{2+} , and Fe^{3+}
$\text{C}_2\text{H}_3\text{O}_2^-$, HCO_3^- , NO_3^- , ClO_3^-	[none]
SO_4^{2-}	Salts of Ag^+ , Ba^{2+} , Ca^{2+} , Hg_2^{2+} , Pb^{2+} , and Sr^{2+}

Ionic compounds display patterns in solubility

Definition

Solubility rules enable us to predict whether a given ionic compound is soluble in water based on the identity of its ions.

Insoluble ionic compounds contain these ions	Exceptions
CO_3^{2-} , CrO_4^{2-} , PO_4^{3-} , S^{2-}	Salts of group 1 cations and NH_4^+
OH^-	Salts of group 1 cations and Ba^{2+}

Using solubility rules, we can predict products

Predicting the Products of Precipitation Reactions

Predict the solid product (if any) when aqueous solutions of the compounds below are combined.

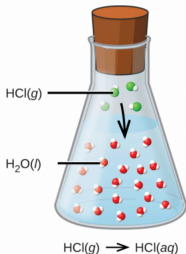
- A. Potassium sulfate and barium nitrate
- B. Lithium chloride and silver acetate
- C. Lead(II) nitrate and ammonium carbonate

Protons are transferred during acid-base reactions

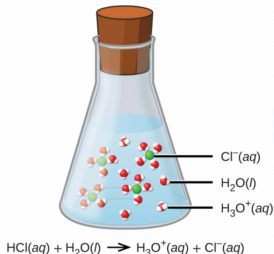
Definition

In an **acid-base reaction**, a hydrogen ion H^+ is transferred from one species (**acid**) to another (**base**). Acids yield hydronium H_3O^+ upon dissolution and reaction in water; bases yield hydroxide OH^- upon dissolution and reaction in water.

Dissolution



Acid-base reaction



Acids and bases can vary in strength

Definition

Strong acids and bases react completely with water to give hydronium or hydroxide (respectively).



Definition

Weak acids and bases react only partially with water.



Acids neutralize bases

Definition

In a **neutralization** reaction, an acid (HX) reacts with a base (MOH) to produce a salt (MX) and water (H_2O).

Acids neutralize bases

Identifying the Acid and Base in a Neutralization Reaction

Label the acid and base on the reactant side of the neutralization reaction of magnesium hydroxide and hydrochloric acid.



Acids react with bases

Writing Chemical Equations for Acid-base Reactions

Write balanced equations for the acid-base reactions below.

- A. The weak acid hydrogen hypochlorite reacts with water
- B. Barium hydroxide reacts with nitric acid in aqueous solution

Electrons are transferred in redox reactions

Definition

In an **oxidation-reduction reaction**, electrons are transferred fully or partially from one species to another. **Oxidation** is the loss of electrons; **reduction** is the gain of electrons.

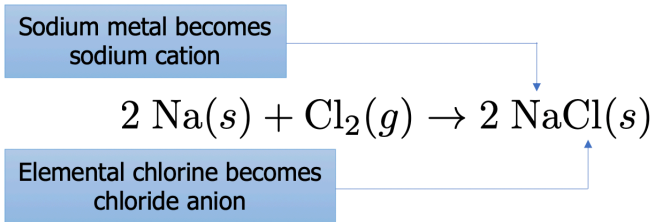


Figure 6: Redox reactions can be split into **half-reactions**.

Oxidation number is a measure of charge

Definition

Oxidation number is the charge that an atom would possess if it were involved in only ionic bonds.

1. The oxidation number of an atom in an element is zero.
2. The oxidation number of a monatomic ion is equal to its charge.
3. Oxidation numbers for common nonmetals are as follows:

Hydrogen -1 when combined with metals; +1 when combined with nonmetals

Oxygen -2 when combined with metals and most non-metals

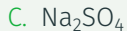
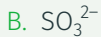
Halogens -1 for fluorine always; + n for others when linked to n oxygens

4. The sum of oxidation numbers for all atoms in a molecule or polyatomic ion equals the charge on the molecule or ion.

Assigning oxidation numbers

Assigning Oxidation Numbers

Assign oxidation numbers to all the elements in the following species.



Redox reactions can take place in acidic solution

Redox reactions in aqueous solution often involve H_2O , H_3O^+ (" H^+ "), or OH^- as reactants or products.

Balancing Aqueous Redox Reactions (Acidic Solutions)

1. Write the two half-reactions representing the redox process.
2. Balance all elements except oxygen and hydrogen.
3. Balance oxygen by adding H_2O molecules.
4. Balance hydrogen by adding H^+ ions.
5. Balance charge by adding electrons.
6. Scale half-reactions to yield equal numbers of electrons in each.
7. Add the balanced half-reactions together and simplify by removing species that appear on both sides of the equation.

Redox reactions can take place in basic solution

Redox reactions in aqueous solution often involve H_2O , H_3O^+ (" H^+ "), or OH^- as reactants or products.

Balancing Aqueous Redox Reactions (Basic Solutions)

1. Balance *as if* the reaction is taking place in acid by following steps 1 – 7 on the last slide.
2. Add OH^- ions to both sides of the equation in numbers equal to the number of H^+ ions.
3. On the side of the equation containing both H^+ and OH^- ions, combine these ions to yield water molecules.
4. Simplify the equation by removing any redundant water molecules.

Balancing redox reactions in acidic solution

Balancing Redox Reactions in Acidic Solution

Write a balanced equation for the reaction between dichromate ion and iron(II) to yield iron(III) and chromium(III) in acidic solution.

Balancing redox reactions in basic solution

Balancing Redox Reactions in Basic Solution

Write a balanced equation for the reaction between dichromate ion and iron(II) to yield iron(III) and chromium(III) in basic solution.

Reaction Stoichiometry

Stoichiometry involves mass and mole relationships

Definition

Stoichiometry refers to the study of quantitative relationships between substances involved in a chemical reaction.

Definition

A **stoichiometric factor** is a ratio of two coefficients in a balanced equation, used to calculate one amount of substance from another.

1 cup mix + 0.75 cup milk + 1 egg $\longrightarrow$ 8
pancakes



Balanced equations are the foundation of stoichiometry

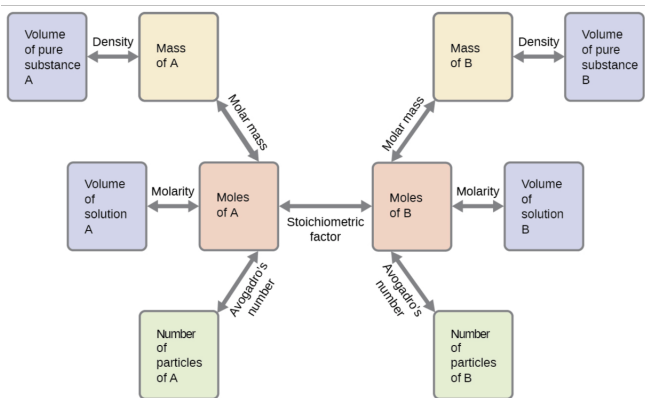


Figure 7: Quantities are related via ratios in the **stoichiometry toolbox** such as molar mass, stoichiometric factors, molarity, etc.

Masses of reactants and products are related

Relating Masses of Reactants and Products

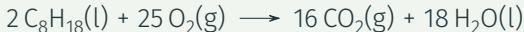
What mass of sodium hydroxide, NaOH, would be required to produce 16 g of magnesium hydroxide, $\text{Mg}(\text{OH})_2$, by the following reaction?



Masses of different reactants are related

Relating Masses of Reactants

What mass of oxygen gas, O_2 , from the air is consumed in the combustion of 702 g of octane, C_8H_{18} , one of the principal components of gasoline?



Reaction Yields

Yields can be limited by reactant amounts

Definition

Stoichiometric amounts are the exact quantities of reactants that combine according to the balanced equation.

1 cup mix + 0.75 cup milk + 1 egg $\longrightarrow$ 8 pancakes

- 2 cups mix
- 3 cups mix
- 1.5 cups mix

Definition

The **limiting reactant** is completely consumed when others are used in excess; it *limits* reaction progress.

Balanced equations are like recipes

Limiting Pancakes

I have 1.5 cups mix, 1.5 cups milk, and 2 eggs. Which ingredient limits the number of pancakes I can make?

1 cup mix + 0.75 cup milk + 1 egg $\longrightarrow$ 8 pancakes

Balanced equations are like recipes

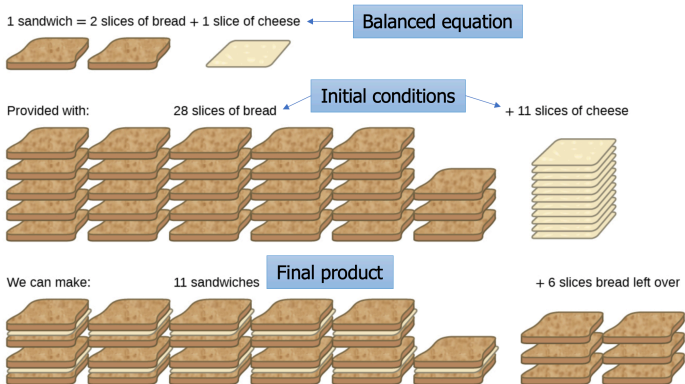


Figure 8: The sandwich analogy for limiting reactant problems.

Reaction events count occurrences of a reaction

Definition

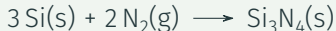
A **reaction event** is a single occurrence of a reaction. Number of reaction events is represented using the variable x (units of mol).

- Imagine "1 reaction event" above every reaction arrow:
$$2 \text{C}_8\text{H}_{18}(\text{l}) + 25 \text{O}_2(\text{g}) \xrightarrow{1} 16 \text{CO}_2(\text{g}) + 18 \text{H}_2\text{O}(\text{l})$$
- The stoichiometric coefficient of A can be read as "moles A consumed/produced for every mole of reaction events"
- Reaction events are a *unifying concept* in stoichiometry
 - Applied in chemical equilibrium and thermodynamics
 - Allow us to determine limiting reactant in a completely generalizable way (calculating x_i for each reactant i)

Amounts of reactants point to a limiting reactant

Determining the Limiting Reactant

Silicon nitride is prepared according to the following equation. Which reactant limits the maximum possible amount of product when 2.00 g of Si and 1.50 g of N₂ react?

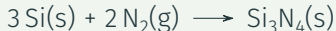


Theoretical yield is based on the limiting reactant

Theoretical Yield

Silicon nitride is prepared according to the following equation.

What is the theoretical yield of Si_3N_4 when 2.00 g of Si and 1.50 g of N_2 react?



Yields are the expected and observed amounts of product

Definition

Theoretical yield is the hypothetical maximum amount of product given amounts of reactants and a balanced equation. **Percent yield** is the ratio of actual yield to theoretical yield expressed as a percentage.

1 cup mix + 0.75 cup milk + 1 egg $\longrightarrow$ 8 pancakes

- 2 cups mix, 1.5 cups milk, 2 eggs

Percent yield is a measure of reaction quality

Calculating Percent Yield

Upon reaction of 1.274 g of copper(II) sulfate with excess zinc metal, 0.392 g copper metal was obtained according to the equation below. What is the percent yield of copper metal?

